Pulkit **Verma**

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Research Interests

AI Safety, AI Planning, Action-Model Learning, Analysis of Abstractions, and Robotics.

Education

Ph.D. in Computer Science, Arizona State University — *Tempe, USA*
Advisor: Prof. Siddharth Srivastava | GPA: 4.0/4.0 — *Exp. Fall 2023*

Master of Technology in Computer Science & Engineering, IIT Guwahati — *Guwahati, India*
Advisor: Prof. P. K. Das | *Thesis*: Resource Usage Analysis for Speech Recognition Techniques | GPA: 9.07/10 — *May 2015*

Bachelor of Engineering in Information Technology, SVITS, Indore — *Indore, India*
Advisor: Prof. Anand Rajawat | *Capstone*: Smart Analyzer - Predict and Cache Webpages based on Usage. — *June 2011*

Publications

Journals and Conferences

Pulkit Verma, Shashank Rao Marpally, and Siddharth Srivastava. "Discovering User-Interpretable Capabilities of Black-Box Planning Agents". In *19th International Conference on Principles of Knowledge Representation and Reasoning* (KR), 2022. (To appear).

Rashmeet Kaur Nayyar*, **Pulkit Verma***, and Siddharth Srivastava. "Differential Assessment of Black-Box AI Agents". In *36th AAAI Conference on Artificial Intelligence*, 2022. *Equal Contribution

Pulkit Verma, Shashank Rao Marpally, and Siddharth Srivastava. "Asking the Right Questions: Learning Interpretable Action Models Through Query Answering". In *35th AAAI Conference on Artificial Intelligence*, 2021.

Pulkit Verma, and Pradip K. Das. "A Comparative Study of Resource Usage for Speaker Recognition Techniques". In *International Conference on Signal Processing and Communication*, 2016.

Pulkit Verma, and Pradip K. Das. "i-Vectors in Speech Processing Applications: A Survey". In *International Journal of Speech Technology*, 18(4), pp.529-546, 2015.

Mayank Gupta, **Pulkit Verma**, Tuhin Bhattacharya, and Pradip K. Das. "A Mobile Agents based Distributed Speech Recognition Engine for Controlling Multiple Robots". In *International Conference on Advances In Robotics*, 2015.

Pulkit Verma, Mayank Gupta, Tuhin Bhattacharya, and Pradip K. Das. "Improving Services using Mobile Agents-based IoT in a Smart City". In *2014 International Conference on Contemporary Computing and Informatics*, 2014.

Workshops, Demos, Posters, Preprints, DCs

Naman Shah*, **Pulkit Verma***, Trevor Angle, and Siddharth Srivastava. "JEDAI: A System for Skill-Aligned Explainable Robot Planning". In *21st International Conference on Autonomous Agents and MultiAgent Systems (Demonstration Track)*, 2022. ***[Best Demonstration Award]***. *Equal Contribution

Pulkit Verma. "Data Efficient Paradigms for Personalized Assessment of Taskable AI Systems". In *ICAPS 2022 Doctoral Consortium* (To appear), and *KR 2022 Doctoral Consortium* (To appear), 2022.

Yizhong Wang et al. "Benchmarking Generalization via In-Context Instructions on 1,600+ Language Tasks". *(In submission)*, 2022.

Pulkit Verma. "Data Efficient Algorithms and Interpretability Requirements for Personalized Assessment of Taskable AI Systems". In *IJCAI 2021 Doctoral Consortium*, 2021.

Pulkit Verma, and Siddharth Srivastava. "Learning Causal Models of Autonomous Agents using Interventions". In *IJCAI 2021 Workshop on Generalization in Planning*, 2021.

Alok Shankar Mysore et al. "Investigating the 'Wisdom of Crowds' at Scale.". In *Adjunct Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology (Poster)*, 2015.

Professional Experience

Arizona State University
Tempe, USA

Graduate Research Associate *Aug. 2018 - (present)*

- Investigating the minimal set of requirements in an AI system that would enable a user to assess and understand the limits of its safe operability.
- Developed a framework which generates interrogation policies to derive an interpretable model of an AI agent.
- Co-lead a team to create an educational tool "JEDAI" to introduce AI planning concepts to laypersons using mobile manipulator robots.
- Co-designed the experiments showcasing possible use-cases of an NSF project on training lay people to use adaptive AI systems which are safe, robust and reliable.

NetApp Inc.
Bengaluru, India

Storage Efficiency Developer *Jul. 2015 - Jul. 2018*

- *Workload Prediction*: Predicted workload on a storage volume based on I/O patterns for All Flash systems. This helped in optimizing data storage and management while keeping the nature of data private to the users.
- *File and Volume Clones (FVCs)*: Developed features and managed issues related to file and volume clones on UNIX-based NetApp proprietary WAFL file system to improve the storage efficiency of enterprise data servers.
- Co-developed a file system component which prevented data loss if FVCs are modified when cloud backup is offline.

Tata Consultancy Services Ltd.
Pune, India

iOS Application Developer *Mar. 2012 - Aug. 2013*

- Built interactive applications for iPhone and iPad and developed tools to analyze the user investment portfolio.
- Performed cross-platform application development of mobile applications supported on both Android and iOS.

Teaching Experience

CSE 571 Artificial Intelligence [Graduate-level Course]
Tempe, USA

Arizona State University *Spring'22, Fall'19*

- Delivered hands-on sessions on Planning Graphs, Markov Decision Processes, Reinforcement Learning, and Neural Networks in Spring 2022.
- Delivered tutorials and readings on Robot Operating System (ROS), Constraint Satisfaction Problems (CSPs), First Order Logic, Planning Domain Definition Language (PDDL), and Dynamic Bayesian Networks in Fall 2019.
- Co-created the software test-bed used to support the ROS-based course projects.
- Created and graded homework assignments and exams.
- Moderated research paper discussions in lectures.
- Mentored project groups working on course projects.

Session Lead [Graduate-level Courses]
Tempe, USA

Arizona State University *2020-2021*

- Moderated three research paper discussions in CSE 574 Planning and Learning Methods in AI, Spring'21 (39 students).
- Delivered tutorial on *Probabilistic Inference* in CSE 471 Introduction to Artificial Intelligence, Fall'20 (118 students).

CS 566 Speech Processing [Graduate-level Course]
Guwahati, India

Indian Institute of Technology Guwahati *Fall 2014*

- Co-developed a custom C++ API to use a voice recording tool that students used in course projects.
- Mentored students working on course projects and homework assignments.
- Graded homework assignments and exams.

Undergraduate Lab. Courses
Guwahati, India

Indian Institute of Technology Guwahati *2013-2015*

- TA for the lab courses: CS 513 Programming Lab.; CS 462 Graphics Lab.; and CS 243 Software Engineering Lab.
- Mentored project groups working on course projects.
- Graded homework assignments and exams.

Service

2022	**Co-organizer**, IJCAI-ECAI 2022 Workshop on Generalization in Planning (GenPlan'22)
2022	**PC Member**, 32nd International Conference on Automated Planning and Scheduling (ICAPS 2022)
2022	**PC Member**, ICAPS 2022 Workshop on Explainable AI Planning (XAIP'22)
2021	**Reviewer**, IEEE Robotics and Automation Letters
2021	**Volunteer**, 30th International Joint Conference on Artificial Intelligence (IJCAI 2021)
2021	**Volunteer**, 31st International Conference on Automated Planning and Scheduling (ICAPS 2021)
2021	**PC Member**, ICAPS 2021 Workshop on Explainable AI Planning (XAIP'21)
2021	**PC Member**, IJCAI 2021 Workshop on Generalization in Planning (GenPlan'21)
2021	**Volunteer**, 9th International Conference on Learning Representations (ICLR 2021)
2021	**Research Grants Reviewer**, Graduate and Professional Student Association, ASU
2021	**GPSA Awards Reviewer**, Graduate and Professional Student Association, ASU
2019	**Volunteer**, Southwest Robotics Symposium 2019, Tempe, USA
2018	**Volunteer**, 16th Int. Conf. on Principles of Knowledge Representation & Reasoning (KR 2018)
2014	**Volunteer**, National Workshop on GPU Programming and Applications, IIT Guwahati

Honors & Awards

May 2022	**Best Demonstration Award**, AAMAS 2022
May 2022	**ICAPS Student Scholarship**, ICAPS 2022
April 2022	**GPSA Teaching Excellence Award**, Arizona State University
March 2022	**SCAI Doctoral Fellowship**, School of Computing and AI, Arizona State University
2021-2022	**GPSA Travel Grant**, Arizona State University (for AAAI 2021, AAAI 2022, IJCAI 2022, KR 2022)
Mar 2021	**Graduate College Travel Awards**, ASU (for AAMAS 2021, ICLR 2021, ICML 2021, IJCAI 2021)
Oct 2018	**Fulton Schools Experiential Learning Grant**, Arizona State University
Aug 2013 - May 2015	**Post Graduation Fellowship**, All India Council for Technical Education, Government of India
June 2012	**TCS ILP Kudos Award**, Tata Consultancy Services Ltd.

Other Affiliations

Jan 2021 - (present)	**Graduate Student**, Center for Human, Artificial Intelligence, and Robot Teaming, ASU
Nov 2020 - (present)	**Affiliated Graduate Student**, Center for Human-Compatible Artificial Intelligence, UCB
Aug 2018 - (present)	**Graduate Student**, Autonomous Agents and Intelligent Robots (AAIR) Lab, ASU
Jul 2016 - Dec 2016	**Online Contributor**, Stanford Scholar Initiative
Jan 2014 - May 2015	**Graduate Student**, Robotics Lab, IIT Guwahati

Mentorship

Trevor Angle
Tempe, USA

M.S. in Computer Science, Arizona State University *2021-2022*

- Co-mentored Trevor at ASU where he worked on a system for skill-aligned explainable robot planning, called JEDAI.
- Resulted in one short paper published in demonstration track at AAMAS 2022.

Shashank Rao Marpally
Tempe, USA

M.S. in Robotics and Autonomous Systems, Arizona State University *2020-2021*

- Mentored Shashank for 3 semesters during his masters where he worked on learning action models for game agents.
- Resulted in two conference publications (AAAI'21 and KR'22).

Masih Tamsoy
Guwahati, India

B.Tech. in Computer Science and Engineering, IIT Guwahati *2014-2015*

- Mentored Masih for 2 semesters at Robotics Lab, IIT Guwahati where he worked on his bachelor's thesis titled "Speaker Verification Systems".
- Helped Masih in developing a speaker verification system using joint factor analysis and i-vectors.